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B.Tech. Degree VI Semester Examination April 2018

ME 15-1606 (E2) ADVANCED MECHANICS OF SOLIDS (2015 Scheme)

Time : 3 Hours

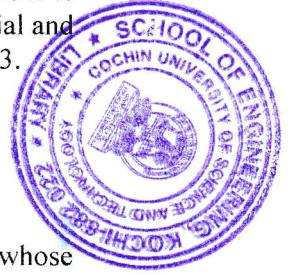
Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Derive the differential equations of equilibrium in 2D Cartesian co-ordinates.
- (b) For a given displacement field $\bar{U} = (x^2y + 5z^2)i + (xzy^2 + y^2)j + x^2y^2k$, determine the strain tensor at (2, -1, 2)
- (c) Obtain the expressions for the radial, tangential and shear strain components in terms of radial displacement u and tangential displacement v .
- (d) A thick cylinder of internal diameter 100 mm and external diameter 200 mm is subjected to an internal pressure of 10 N/mm². Find the variation of the radial and circumferential stresses across the wall thickness, if $E = 200$ GPa and $\nu = 0.3$.
- (e) Obtain the strain compatibility conditions for 3D system.
- (f) Derive the Cauchy's stress formula $T_i = \sigma_{ij}n_j$.
- (g) Given $\sigma_{ij} = \begin{bmatrix} 4 & 3 & -2 \\ 3 & 2 & 8 \\ -2 & 8 & -2 \end{bmatrix}$, calculate the stress traction vector on a plane whose normal has direction cosines $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$.
- (h) State the Castigliano's theorems.
- (i) Explain how Prandtl's membrane analogy can be used to analyse the torsion of non-circular bars.
- (j) What is the importance of shear centre? Explain.



PART B

(4 × 10 = 40)

- II. Given the following stress field in a body $\sigma_{xx} = 80x^3 + y$, $\sigma_{xy} = 100(y^2 + 1)$,
 $\sigma_{yy} = 100(x^3 + 10)$, $\sigma_{yz} = 0$, $\sigma_{zx} = x(z^3 + 100xy)$, $\sigma_{zz} = 10(10z^3 + 9y^2)$,
 find the body force distribution necessary to satisfy the equilibrium. What are the stresses and body forces at point (1, 1, 5)?

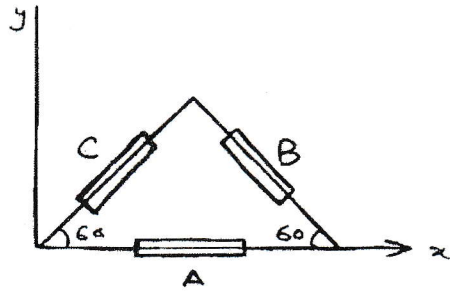
OR

(P.T.O.)

- III. A 60° delta strain rosette consisting of three electrical resistance strain gauge arranged as shown. If the readings are :

$\epsilon_a = 10 \times 10^{-6}$, $\epsilon_b = -5 \times 10^{-6}$, $\epsilon_c = 4 \times 10^{-6}$, compute :

- The principal strains ϵ_1, ϵ_2
- Maximum shear strain $\gamma_{\max}$
- Principal stresses σ_1, σ_2
- Maximum in-plane shear stress.



- IV. For an axi-symmetric stress distribution, neglecting body forces show that the differential equation of equilibrium can be reduced to the form

$$\frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr} - \frac{u}{r^2} = 0, \text{ where } u \text{ is the radial displacement. Solve this equation}$$

and obtain the solution $u = C_1 r + \frac{C_2}{r}$.

OR

- V. A steel shaft of 100 mm diameter has a 500 mm steel disc shrunk on it. The difference in diameters of the shaft and the disc at the interface is 0.08 mm. Find the radial and tangential stresses in the assembly due to the shrink fit. Also calculate the speed at which the disc will be loosened from the shaft. Take $E = 200 \text{ GPa}$, $\nu = 0.3$, $\rho = 8000 \text{ kg/m}^3$

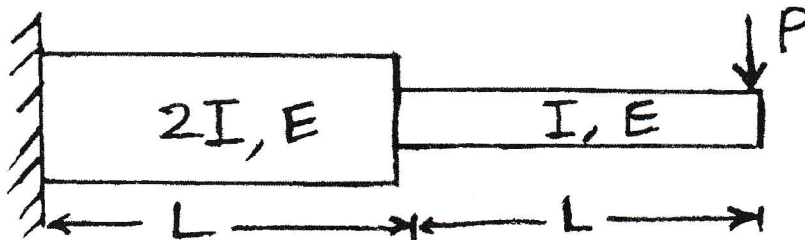
- VI. The state of stress at a point in a material is given by

$$\sigma_{ij} = \begin{bmatrix} 70 & -40 & 20 \\ -40 & 10 & 20 \\ 20 & 20 & -20 \end{bmatrix} \text{ MPa. Determine the principal stresses, maximum shear}$$

stresses, and the orientation of the principal planes.

OR

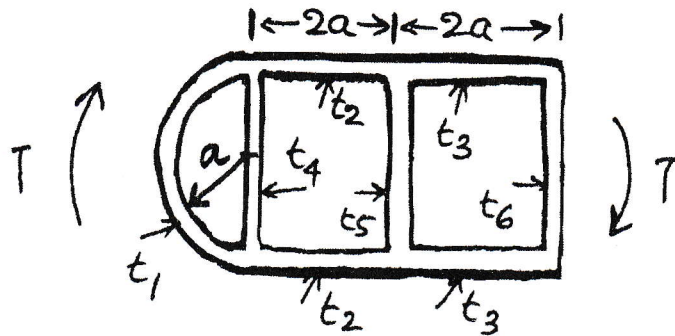
- VII. A cantilever beam is loaded as shown. Using Castigliano's theorem, find the slope and deflection at the free end.



(Contd...3)

VIII.

The figure shows a tubular section with three cells. The thin walled tube is subjected to a torque $T = 113 \text{ kN.m}$. Determine the shear stresses in the walls of the section.



$$a = 12.7 \text{ cm}$$

$$t_1 = 0.06 \text{ cm}$$

$$t_2 = 0.08 \text{ cm}$$

$$t_3 = 0.08 \text{ cm}$$

$$t_4 = 0.13 \text{ cm}$$

$$t_5 = 0.08 \text{ cm}$$

$$t_6 = 0.10 \text{ cm}$$

OR

IX.

A cantilever beam of equal leg angle section of $100 \text{ mm} \times 100 \text{ mm} \times 1 \text{ m}$ is subjected to a load of 3 kN at the free end acting at 20° to the vertical. Calculate the bending stress at A and also the position of neutral axis.

